Eleanor K. MacDonald

Sewall Conservation Ecology Lab, Department of Biology, Temple University
eleanor.macdonald@temple.edu

Education

Temple University, Department of Biology - Doctorate of Philosophy in *Biology* **2026 – Present**

California Polytechnic State University, Department of Wildlife Humboldt - Master of Science in *Natural Resources with emphasis in Wildlife* **2024 – 2025**

The Ohio State University, Department of Evolution, Ecology & Organismal Biology - Bachelor of Science in *Evolution & Ecology with Research Distinction* **2020 – 2022**

The Ohio State University, John Glenn College of Public Affairs - Bachelor of Arts in *Public Management, Leadership, & Policy,* Minor in *Environmental Science* **2015 - 2018**

Research Experience

Temple University, Philadelphia, Pennsylvania **2026 – Present**
Graduate Researcher, Sewall Conservation Ecology Lab
- Developing dissertation research examining how environmental change and conservation management shape behavioral and demographic responses in butterfly and bird populations and communities

University of Illinois, Urbana-Champaign, Champaign, Illinois **2026**
Research Associate, Ward Lab
- Designed and conducted randomized field experiments comparing automated nocturnal insect camera traps (Mothbox and Moth-o-Matics) and UV bucket traps for sampling nocturnal insect communities
- Developed R computer vision workflows to estimate moth biomass from trap imagery and validated estimates against physical specimen dry mass

California Polytechnic State University, Arcata, California **2024 – 2026**
Graduate Researcher & Lab Manager, Johnson Habitat Ecology Lab
- Managed multi-year field research across 20 Napa Valley vineyards, conducting over 2,000 songbird nest box evaluations to collect avian survival, morphometric, and fecal DNA samples; fitted Bayesian hierarchical N-mixture models in R to evaluate avian pest control services and managed laboratory administrative operations and personnel
- Engaged diverse stakeholders through outreach talks and webinars, from local to international audiences, highlighting the ecological benefits of nest boxes and supporting conservation engagement

The Ohio State University Aquatic Ecology Lab, Columbus, Ohio **2022 – 2023**
Senior Research Assistant, Hood Lab
- Supervised seven undergraduate students to count, identify, and measure invasive zooplankton in Lake Erie aquatic samples as well as conduct spectrophotometry of chlorophyll samples
- Identified, counted, and measured zooplankton communities in aquatic samples from Lake Erie and conducted spectrophotometry of chlorophyll samples

The Ohio State University, Columbus, Ohio **2020 – 2022**
Student Researcher, Goodell Lab
- Conducted and defended undergraduate thesis research: Patterns of Taxonomic and Functional Diversity of a Group of Non-bee Pollinators (Diptera: Syrphidae) in Ohio
- Completed plant phenology annotations on iNaturalist to track flowering and fruiting times for a specialist bee guide

Teaching Experience

Teaching Assistant **2026**
Temple University, Introduction to Organismal Biology
- Leads laboratory instruction for 40 undergraduate students in organismal biology concepts, techniques, and experimental procedures

Teaching Assistant **2025**
California Polytechnic State University, Wildlife Diseases
- Supported undergraduate learning in wildlife disease and ecology by assisting with labs and grading assignments

Teaching Assistant **2024**
Cal Poly Humboldt, Non-game Wildlife Management: Shorebird Ecology and Conservation
- Supported undergraduate instruction in shorebird identification and ecology, including migration ecology concepts and shorebird conservation, through lab and field trip supervision and grading assignments

Teaching Assistant **2021**
The Ohio State University, Behavioral Ecology
- Designed course material and provided instruction on experimental research design, literature review, research ethics, ArcGIS and elementary statistical analyses
- Supervised independent research opportunities in behavioral ecology for undergraduates considering careers in basic scientific research, secondary education, or graduate research programs in biological and behavioral sciences

Laboratory Teaching Assistant **2020 – 2021**
The Ohio State University, General Chemistry
- Facilitated undergraduate chemistry labs by preparing experiments, monitoring labs, and addressing student questions

Publications

Jomaa, F.W., C. Pham, **E.K. MacDonald**, B. Martinico, D.S. Karp, M.D. Johnson. 2026. Public information is used for prospecting but not ultimate nest site selection by Western Bluebirds (*Sialia mexicana*). *Ecology and Evolution.*

Adler, K., B.L. Martinico, **E. MacDonald**, S.M. Kross, D.S. Karp, A. Turner, and M.D. Johnson. 2024. Unraveling the influence of uncultivated habitat on avian grape damage in a viticultural landscape. *Proceedings of the 31st Vertebrate Pest Control Conference.*

In Review:

MacDonald E.K., A. Turner, R. Hoey-Chamberlain, F. Jomma, K. Gallardo-Cruz, B.L. Martinico, D.S. Karp, E.E. Wilson-Rankin, H. Wilson, M.D. Johnson. Nest Site Limitation Drives Spatial Structure of Avian-Mediated Arthropod Control in an Agroecosystem. In review at *Ecology Letters.*

MacDonald E.K., M. Fernandez-Duque, J.J. Falk, A. Romero, E. Fernandez-Duque, K. Kochvar, M.E. Hauber, F. Fernandez-Duque. Consistent arrival sequences at feeders reveal species-level producer-scrounger dynamics in a wild hummingbird community. In review at *Behavioral Ecology.*

MacDonald E.K., A. Turner, D.S. Karp, H. Wilson, B.L. Martinico, E.E. Wilson-Rankin, M.D. Johnson. Local habitat heterogeneity supports insectivorous birds and arthropod diversity in a vineyard ecosystem. In review at *PLoS One.*

MacDonald E.K., J. Nehl, D.S. Karp, H. Wilson, B.L. Martinico, E.E. Wilson-Rankin, M.D. Johnson. Temporal dynamics structure lady beetle communities while cover crops and semi-natural habitat support key taxa in organically managed vineyards. In review at *Biological Control.*

MacDonald, E.K., M. Spring, K. Goodell. Landscape cover and habitat proximity as drivers of hover fly (Diptera: Syrphidae) abundance and diversity. In review at *Environmental Entomology.*

Karp, D.S., J.A. Smith, C. Pham, K.V. Gallardo Cruz, F. Jomaa, **E.K. MacDonald**, B. Martinico, E.E. Wilson-Rankin, M.D. Johnson. Avian Behavioral Thermoregulation to Hot Temperatures Carries Reproductive Opportunity Costs in Agriculture. In review at *Ecology.*

Awards, Grants, and Fellowships

Honors & Awards
Koplin Award for Outstanding Wildlife Graduate Student, Department of Wildlife, Cal Poly Humboldt, 2025
National Science Foundation Graduate Research Fellowship, Honorable Mention, 2024

Grants Awarded
Wildlife Graduate Student Society Grant, 2025

American Ornithological Society Travel Grant, 2024
OSU College of Arts and Sciences Undergraduate Student Research Scholarship, 2021
OSU Undergraduate Student Research Grant, 2020

Fellowships
Endangered Species Coalition, *Activist Training Lab Fellow,* 2020 – 2021
Washington Academic Internship Program, *John Glenn Fellow,* 2018

Presentations

MacDonald E., A. Turner, R. Hoey-Chamberlain, F. Jomma, K. Gallardo-Cruz, B.L. Martinico, D.S. Karp, E.E. Wilson-Rankin, H. Wilson, M.D. Johnson. 2026. Habitat mediates pest control provided by birds using nest boxes in winegrape vineyards. Landscape Management for Functional Biodiversity Conference. Oral Presentation.

MacDonald, E. & M.D. Johnson. 2025. Avian Ecosystem Services in Vineyards. Wild Farm Alliance. Webinar offered as continuing education credit for the California Department of Pesticide Regulation.

MacDonald, E., A. Turner, R. Hoey-Chamberlain, D.S. Karp, E.E. Wilson-Rankin, H. Wilson, M.D. Johnson. 2025. Evaluating Avian-Insect Interactions and Ecosystem Services in Vineyards. American Ornithological Society (AOS) Conference. Oral Presentation.

MacDonald, E. 2025. Climate Talks: Natural Predators in Vineyards**.** The Regenerative Viticulture Foundation and The Porto Protocol. Webinar Panelist.

MacDonald, E. 2025. What do birds have to do with wine? Shasta Birding Society. Oral Presentation.

MacDonald, E., A. Turner, C. Pham, D.S. Karp, M.D. Johnson. 2025. Examining the effects of songbird nest boxes and land use on avian community composition and functional diversity in Napa Valley vineyards. The Wildlife Society Western Section Conference. Oral Presentation.

MacDonald, E., A. Turner, C. Pham, D.S. Karp, M.D. Johnson. 2024. Examining the effects of songbird nest boxes and land use on avian community composition and functional diversity in Napa Valley vineyards. Agricultural Research Institute Conference. Poster.

MacDonald, E., A. Turner, C. Pham, D.S. Karp, M.D. Johnson. 2024. Examining the effects of songbird nest boxes and land use on avian community composition and functional diversity in Napa Valley vineyards. American Ornithological Society (AOS) Conference. Poster.

MacDonald, E., M. Spring, K. Goodell. 2023. Patterns of Taxonomic and Ecological Diversity of a Non-bee Pollinator (Diptera: Syrphidae) in Ohio. Ohio Natural History Conference. Poster.

MacDonald, E., M. Spring, K. Goodell. 2022. The use of bycatch and community science to measure biodiversity: A statewide survey of Ohio Hover Flies (Diptera: Syrphidae). North Central Branch Virtual Meeting for the Entomological Society of America. Poster. **Award received: Second place recipient.**

MacDonald, E., M. Spring, K. Goodell. 2022. Patterns of Taxonomic and Ecological Diversity of a Non-bee Pollinator (Diptera: Syrphidae) in Ohio. Ohio State University Denman Undergraduate Research Forum: Animal and Insect Sciences. Poster. **Award received: First place recipient.**

MacDonald, E., M. Spring, K. Goodell. 2021. Geographic and ecological patterns of hover fly (Diptera Syrphidae) diversity in Ohio. Ohio State University at Newark Virtual Student Research Forum. Oral Presentation.

MacDonald, E., M. Spring, K. Goodell. 2021. Patterns of hover fly (Diptera Syrphidae) diversity in Ohio: Do hover fly abundances track bee abundances across the state? North Central Branch Virtual Meeting for the Entomological Society of America. Poster. **Award received: Third place recipient.**